1000+ Synthetic Benchmark Problems for Methods in Dynamic Modeling (Project C03)

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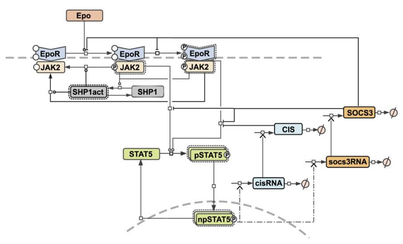
Background

Dynamical Modeling in Systems Biology

(A) Signaling Pathway Diagram [3]

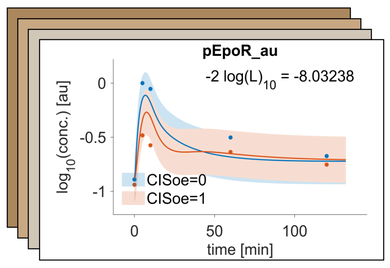


(B) ODE Model

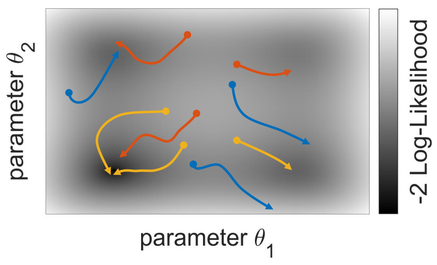
$$\dot{x}_1(t) = f_1(x_1(t), \dots, x_n(t), \vec{u}(t), \vec{\theta})$$
$$\vdots$$
$$\dot{x}_n(t) = f_n(x_1(t), \dots, x_n(t), \vec{u}(t), \vec{\theta})$$

concentrations $\vec{x}$, inputs $\vec{u}$, parameters $\vec{\theta}$

(C) Experimental Data



(D) Numerical Parameter Estimation



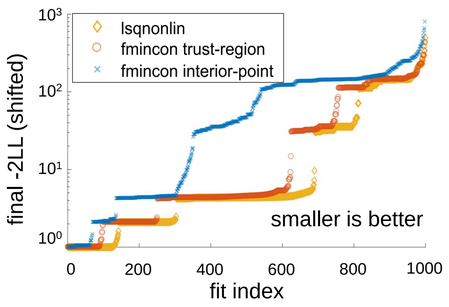
The goal of Systems Biology is to understand the mechanism of biological systems (e.g. in intracellular signaling pathways (A)). The interactions of the involved biomolecules (e.g. proteins, RNA) can be translated into ordinary differential equations (ODEs).

The model parameters can be estimated from experimental data by numerically optimizing the log-likelihood function (C,D). There are different ODE solvers and optimizers, each with multiple hyperparameters. In practice, these choices have to be made by the researcher. But is there a general rule, which method works best (in which situation)?

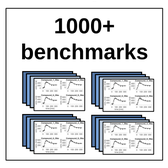
Some method comparisons (E) have already been performed [1,2]. But for comprehensive benchmarking a large set of problems is required (F), and ODE models with published experimental data are rare. We solve this *small data challenge* with synthetic data generation.

Finding the Best Method

(E) Comparing Different Optimizers [1]



(F) We require:



Large Set of Benchmark Problems

For testing and comparing:
- different optimizers
- different ODE solvers
- different hyperparameters

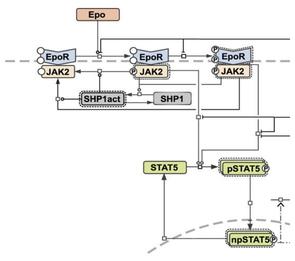
Methods

Generate novel ODE modeling problems

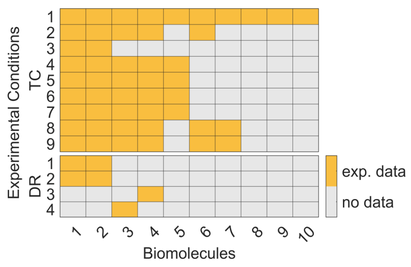
They should be realistic and add variablity to existing problem set.

20 Published ODE Models as Simulation Templates [1]

ODE System



Template Specific Data Structure



Typical Data Structure:
- Time-course (TC) and dose-response (DR) data
- Multiple experimental conditions (e.g. celltypes, knockouts, external stimuli)
- Experimental constraints → partially observed
- Hard to generate from scratch

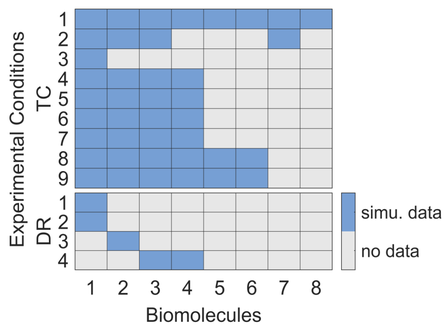
1. **Modify Model Dynamics**
   Multiply model parameters by factors ~ log2uniform(-1, 1)

Simulating Realistic Synthetic Benchmark Problems

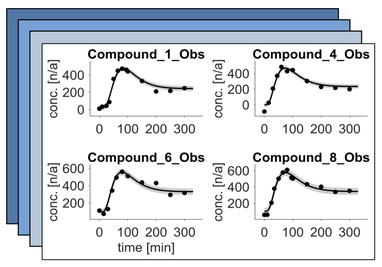
2. **Which Molecules are Observed? [4]**
   How many?
   Individually or jointly?
   Transformations (scale, offset, log10)?

3. **Generate Data Structure**
   Draw columns of template

Synthetic Data Structure



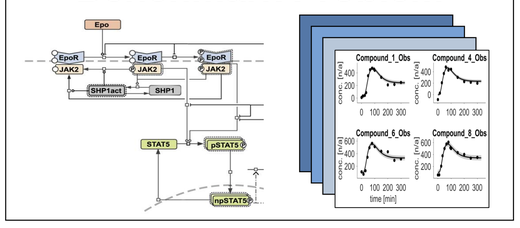
4. **Generate Synthetic Data [4]**
   Characterize time-scales → time-points
   Model + realistic noise → synth. measurements



5. **Export in Standardized Formats [5]**

SBML PEtab

Realistic Synthetic Benchmark Problem



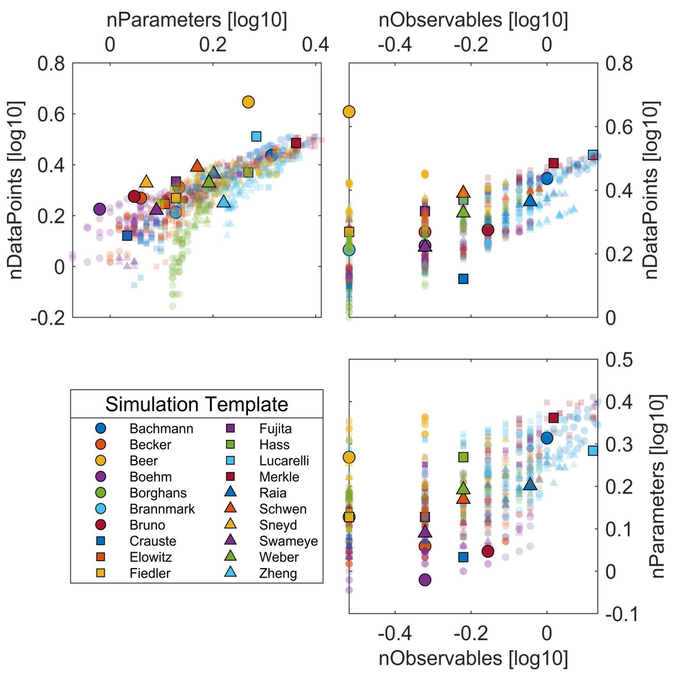
Results & Conclusions

Characteristics of 20 Simulation Templates [1] | **Characteristics of the 50 Realistic Simulations Each**

Template	nStates	nConds	nObs	nParams	nDataPoints
1. Bachmann	25	23	4 – 23	49 – 323	237 – 1714
2. Becker	6	1	1 – 6	10 – 26	11 – 78
3. Beer	4	19	1 – 3	89 – 203	197 – 707
⋮	⋮	⋮	⋮	⋮	⋮
20.Zheng	15	1	2 – 15	48 – 82	21 – 161
Summary	**3 – 33**	**1 – 61**	**1 – 23**	**7 – 376**	**5 – 1714**

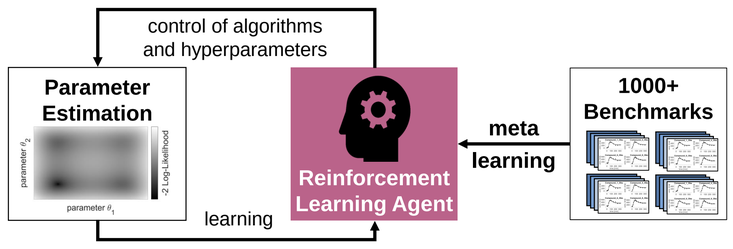
We present 1000 synthetic benchmark problems for dynamic modeling.
They largly increase the number and variablity of the existing problem set, while staying in a range of realistic problem characteristics.

Benchmark Problem Characteristics



Outlook:

This new set of 1000 benchmark problems allows comprehensive testing and comparison of ODE modeling methods. We will investigate different ODE solvers, optimizers and their hyper-parameters.



Project C03 aims at „learning“ the best optimization method. Training on this large set of problems will allow to find a good default policy through „Meta Learning“.

References

[1] Hass et Al (2019). Benchmark problems for dynamic modeling of intracellular processes. Bioinformatics, 35(17), 3073-3082.
[2] Städte et Al (2021). Benchmarking of numerical integration methods for ODE models of biological systems. Scientific reports, 11(1), 2696.
[3] Bachmann et Al (2011). Division of labor by dual feedback regulators controls JAK2/STAT5 signaling over broad ligand range. Molecular systems biology, 7(1), 516.
[4] Egert & Kreutz (2023). Realistic simulation of time-course measurements in systems biology. Mathematical Biosciences and Engineering, 20(6), 10570-10589.
[5] Schmiester et Al (2021). PEtab—Interoperable specification of parameter estimation problems in systems biology. PLoS computational biology, 17(1), e1008646.